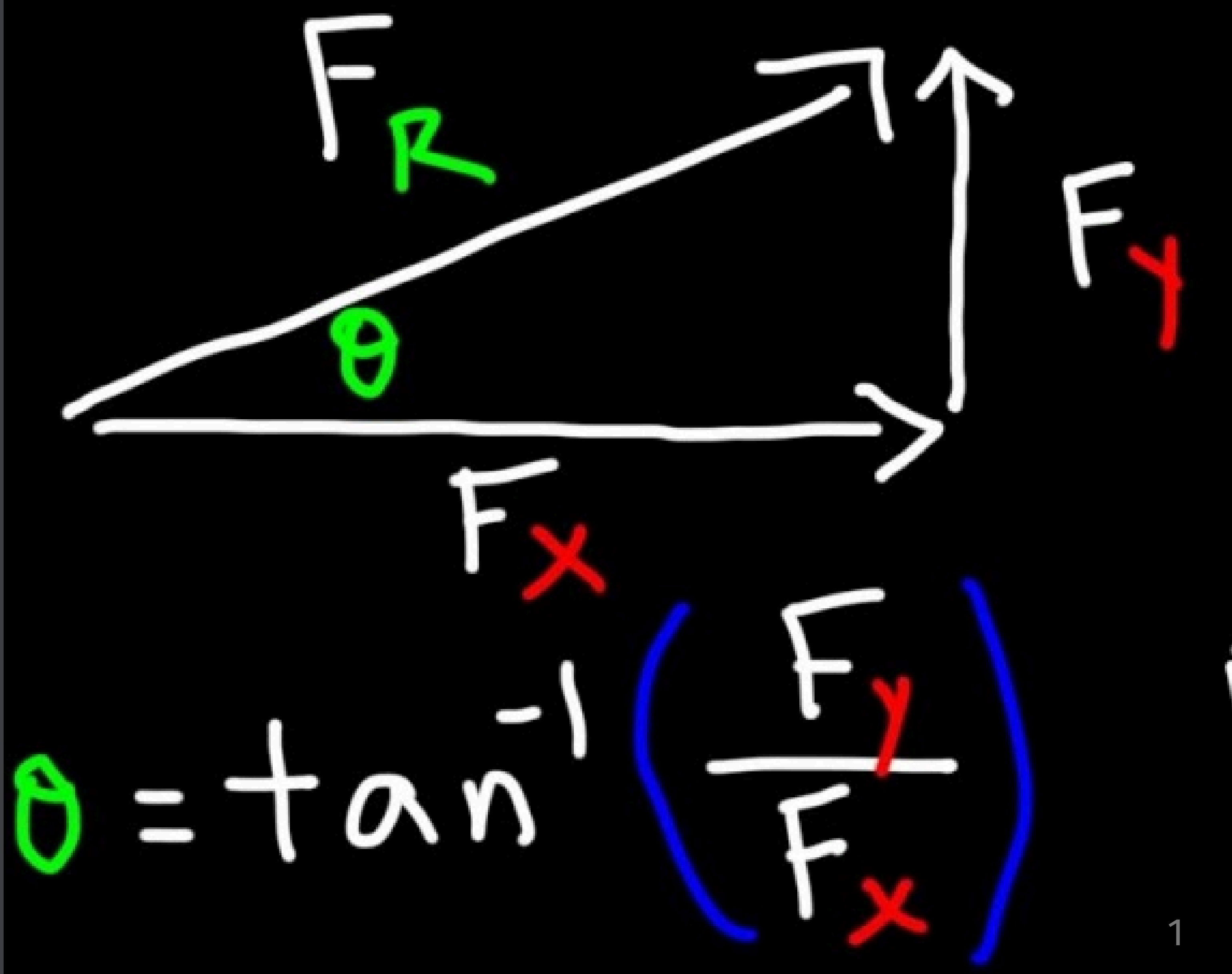
Day 04 - Vector Reminders



Announcements

- Homework 1 is due this Friday
- Homework 2 is posted now
- Help sessions start this week
    - DC today at 4pm; Friday at 3pm (1248 BPS)

Seminars this week (Tuesday and Wednesday)

TUESDAY, January 21, 2025

- **High Energy Physics Theory Seminar**
  - 11:00am, FRIB 1200 lab; Speaker: **Alexei Bazavov**, MSU-CMSE/PA
  - Title: *Lattice QCD: From classical computation to quantum simulation*

WEDNESDAY, January 22, 2025

- **Astronomy Seminar**
  - 1:30 pm, 1400 BPS; Speaker: **Allyson Bieryla**, CfA | Harvard & Smithsonian
  - Title: *Exoplanets and Solar Eclipses for Research and Community Engagement*

Seminars this week (Wednesday, cont.)

- **PER Seminar**
  - 3:00 pm., BPS 1400; Speaker: **Justin Gambrell**, Assistant Professor, Department of Computational Mathematics, Science, and Engineering, Michigan State University - **MSU PA ALUMNUS**
  - Title: *Computational Thinking Assessment for Introductory Physics: Design, Implementation, and Future Directions*
- **FRIB Nuclear Science Seminar**
  - 3:30pm., FRIB 1300 Auditorium; Speaker: Calem Hoffman of Argonne National Laboratory
  - Title: The Influence of Near-Threshold States on Nuclear Observables

Goals for this week

Be able to answer the following questions.

- What are the essential physics models for single particles?
- How do we setup problems in classical mechanics?
- What mathematics do we need to get started?
- How do we solve the equations of motion?

Reminders from Day 03

- In a Newtonian world, we start from a vector description of motion
- Differential equations are mathematical models that describe the motion of particles
- We can use different methods to solve these differential equations

Clicker Question 4-1

Consider the generic position vector $\vec{R}$ for a particle in 2D space. Which of the following describes the direction of the vector in plane polar coordinates (r, ϕ)?

1. $\hat{R}$
2. $\hat{r}$
3. $\hat{\phi}$
4. Some combination of $\hat{r}$ and $\hat{\phi}$
5. I'm not sure.

Group Discussion 4-1

Consider the generic position vector $\vec{R}$ for a particle in 2D space. Find the velocity vector $\vec{V}$ for the particle in Cartesian coordinates (x, y).

What happens in plane polar coordinates?

$$\vec{R} = r\hat{r} + \phi\hat{\phi}$$

Note:

$$\hat{r} = \cos(\phi)\hat{x} + \sin(\phi)\hat{y}$$

$$\hat{\phi} = -\sin(\phi)\hat{x} + \cos(\phi)\hat{y}$$